PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Teapot-Your best intelligence business partner

Version: 2.0

1. Project Overview

- **Problem Statement:** Starting up business often seems overwhelming for most people when it comes to margins, markets pain points in specific niche, business plan, business models, and to what extent the business idea is realistic. This stops a lot of amazing ideas being buried under fear or even due to bad money management. Though current ai aids with strong computing power which leads to super speed data analysis, it is still limited at the theory level. Unless there are people with strong technical knowledge who take parts in second level comprehension of the ai output, or else the information explosion will make most people be scared of taking one more step forward.

- **Target Domain:** enable business starters for small to medium scale businesses and judge based on realistic scenario. Aids business to expand and modify ideas with supported data.

- **Proposed Solution Summary:** Teapot is an AI-powered business intelligence platform built with FastAPI (backend) and Next.js (frontend) that enables business decision-makers to leverage specialized AI agents for strategic insights. The platform features six intelligent agents—Decision Recommendation, Pain Point Analyzer, Profit Optimizer, Resource Optimizer, Risk Identifier, Sales Predictor, and Scenario Simulator—each powered by LLMs like Google Gemini, Anthropic Claude, and GLM like ilmu-glm-5.1 to provide data-driven business recommendations. It includes comprehensive user management with OAuth 2.0 social login (GitHub/Google), role-based admin controls, a real-time chat interface for consulting with agents, project management capabilities, and document generation (PDF/PowerPoint). The system is built on PostgreSQL, uses encrypted API key

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storage, JWT authentication, and provides detailed admin dashboards for user/agent/model configuration and token usage tracking.

2. Background & Business Objective

- **Background of the Problem:** Starting a business has been an abstract topic for many years, as it is a very big topic that could have incredibly different outcome due to a single different twitch in decision, idea, new products, or even words. This causes business doesn't have a really 'start up guide' to follow, we only learn principles. Principle only stays in theory and having different interpretation in everyone, this making some people looks like they are "born for business" or some look like they are "unlucky in start-ups.

- **Importance of Solving This Issue:** While ideas are the precious key of a new business, we do not hope that any diamonds are concealed under sand. A good idea could be something more than bringing wealth to individuals but also bringing changes to the world. Just like the idea of LLMs and GLMs, if the first person having the idea were fear of stepping out to start develop, the data analyst might still need to holding stacks of paper, facing uncountable lines of excel trying to figure out the data they need, the designer might still need to take a one whole week just to make some tweaks on the design due to client's need.
- **Strategic Fit / Impact:** The model interprets user's data and validates their idea based on current market, separation of agent making each agent able to work on more specific niche such as pain point analyzing, and scenario simulating. More than just in theory, the agents are capable of generating actual plans to execute with details.

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3. Product Purpose

3.1. Main Goal of the System: Bridge the gap between abstract business theory and real-world execution by empowering entrepreneurs to transform ideas into actionable, data-driven, and realistic strategies through AI-driven decision support.

3.2. Intended Users (Target Audience):

- Entrepreneurs
- Small-to-Medium Sized Business Owner
- Freelancers
- Corporate innovation teams (intrapreneurs)

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4. System Functionalities

4.1. Description: The system provides comprehensive business assistance and consultation powered by advanced AI models and specialized agents that support users in decision-making, idea generation, and document review. By analyzing user inputs alongside relevant market data, it delivers data-driven insights to evaluate feasibility, identify risks, and uncover opportunities. Through an interactive interface, the platform enables users to refine strategies and transform ideas into practical, executable business plans.

4.2. Key Functionalities:

- **Multi-Agent AI Consultation:** The platform integrates multiple specialized AI agents (e.g., decision recommendation, risk analysis, profit optimization) to provide focused and comprehensive business insights.
- **Document Review and Generation:** The platform assists in reviewing, improving, and generating business documents such as proposals, plans, and presentations.
- **Data-Driven Decision Support:** The system analyzes user input and relevant data to deliver actionable recommendations, reducing reliance on guesswork and intuition.
- **Project and Strategy Management:** Users can manage multiple business ideas or projects, track progress, and refine strategies over time.
- **Admin Dashboard and System Control:** Administrators can manage users, configure AI models/agents, and monitor system usage such as token consumption.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection

The proposed system leverages multiple large language models (LLMs), including Z.AI GLM, Google Gemini, and Anthropic Claude, to power its AI-driven business assistance capabilities. Among these, Z.AI GLM is selected as a core model due to its strong performance in structured reasoning, cost efficiency, and adaptability to diverse business-related tasks. Its ability to interpret complex multi-step instructions, generate structured long-form outputs (such as business plans and analysis reports), and maintain contextual consistency across long conversations makes it particularly suitable for tasks like market evaluation, idea validation, and scenario-based planning. These capabilities align closely with the system's objective of transforming abstract business concepts into actionable and realistic strategies.

Z.AI GLM is especially effective in supporting the platform's multi-agent architecture, where each agent focuses on a specific domain such as risk identification, profit optimization, or decision recommendation. The model's strength in following strict prompt hierarchies and producing well-formatted, segmented responses allows it to generate consistent outputs that can be directly used by downstream modules without heavy post-processing. This is particularly important for maintaining reliability when synthesizing insights from multiple agents into a unified business recommendation. Furthermore, its efficiency in token usage and fast inference speed makes it a practical choice for scalable deployment within a web-based environment, ensuring responsive user interaction even under high demand.

In addition to Z.AI GLM, models such as Google Gemini and Anthropic Claude are integrated to complement the system with their respective strengths, including advanced reasoning, conversational fluency, and safety alignment. By combining these models, the system achieves a balanced approach that enhances accuracy, depth of analysis, and user experience. Overall, the selection of Z.AI GLM as a primary LLM, supported by

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other advanced models, ensures that the platform can deliver comprehensive, reliable, and efficient business intelligence tailored to the needs of modern entrepreneurs.

4.3.2. **Prompting Strategy**

The system adopts a multi-step agentic prompting strategy, where complex business problems are decomposed into smaller, domain-specific tasks handled by different AI agents. Instead of relying solely on zero-shot or few-shot prompting, each agent is designed with structured role-based instructions, context-aware prompts, and iterative reasoning steps to simulate a specialized consultant (e.g., risk analyst, profit optimizer, or decision advisor). This approach allows the system to progressively refine outputs, where intermediate results from one agent are passed as inputs to another, enabling deeper analysis and more reliable conclusions.

This strategy is well-suited to the use case because business decision-making is inherently multi-faceted and cannot be effectively addressed through a single-pass response. Entrepreneurs often require layered insights, such as understanding market feasibility, identifying risks, optimizing resources, and evaluating different scenarios simultaneously. A multi-step agentic approach ensures that each aspect is analyzed in depth before being integrated into a final recommendation, reducing oversimplification and improving accuracy. Additionally, it aligns with the platform's goal of providing structured, actionable business intelligence rather than generic advice, making the system more practical and trustworthy for real-world decision support.

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4.3.3. **Context & Input Handling**

The system handles unstructured and oversized inputs through preprocessing, chunking, and hierarchical summarization to ensure important information is retained while staying within LLM context limits. Since users may submit long business documents such as plans, reports, or research notes, the system first normalizes and structures the input by identifying key sections and removing unnecessary noise. The maximum input size is constrained by the context window of the underlying LLMs (typically several thousand tokens), and inputs exceeding this limit are not rejected outright but are processed using a chunking strategy.

When the input is too large, it is split into smaller, semantically meaningful segments based on paragraphs or topic boundaries. Each segment is analysed separately, and the results are summarized before being combined into a unified context for downstream AI agents. For extremely large or unstructured inputs, a hierarchical summarization process is applied to progressively condense the information.

If the input is overly noisy or unprocessable, the system may prompt the user to refine or restructure their input. This ensures accurate, efficient, and reliable AI-driven analysis.

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4.3.4. **Fallback & Failure Behavior:**

When the model returns off-topic, low-quality, or potentially hallucinated responses, the system applies a multi-layer fallback and validation mechanism to maintain reliability. First, each AI agent output is checked through a lightweight post-processing layer that evaluates response relevance against the original user prompt and expected output format. If the response is detected as off-topic, incomplete, or inconsistent, the system triggers an automatic retry using an adjusted prompt that includes clearer constraints, reinforced context, and stricter output formatting rules. This retries mechanism helps the model self-correct without requiring user intervention.

If repeated attempts still produce unusable outputs, the system falls back to an alternative LLM (for example, switching between Z.AI GLM, Gemini, or Claude) to leverage different reasoning strengths and reduce model-specific hallucination risks. In addition, critical outputs such as business recommendations or risk analysis are passed through a validation layer that checks for logical consistency and alignment with provided inputs.

For cases where confidence remains low after retries and model switching, the system enters a graceful degradation state. Instead of generating potentially misleading content, it provides a structured partial response indicating uncertainty and suggests refining the input. While there is no direct human escalation path in the current hackathon prototype, the architecture is designed to support future integration of human-in-the-loop review for high-stakes decisions.

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5. User Stories & Use Cases

- **User Stories:**
  - As an aspiring entrepreneur, I want to describe my business idea in natural language so that I can receive structured, data-driven business strategies without needing prior technical or business expertise.
  - As a startup founder, I want to analyse risks, profits, and market feasibility so that I can make more confident and informed decisions.
  - As a business student, I want to simulate different business scenarios so that I can understand real-world decision-making outcomes in a practical way.
  - As a small business owner, I want to generate and refine business documents such as proposals and plans so that I can save time and improve quality.

- **Use Cases (Main Interactions):**
  - User inputs a natural language prompt describing a business idea or problem.
  - The system parses the intent and extracts key business components (market, product, risk, cost, etc.).
  - Relevant AI agents are triggered (e.g., Pain Point Analyzer, Profit Optimizer, Risk Identifier).
  - The system generates a structured output such as feasibility report, strategy suggestions, and recommendations.

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- **Use Case Diagram:**

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6. Features Included (Scope Definition)

- A conversational AI interface that allows users to input business ideas and receive structured business analysis through multi-agent consultation.
- A centralized project workspace where users can manage multiple business ideas, track progress, and revisit AI-generated insights.
- Role-based admin dashboard for managing users, AI agents, models, and monitoring token usage.
- Real-time AI consultation via chat, enabling iterative refinement of ideas and strategies.

7. Features Not Included (Scope Control)

- No real-world financial transaction handling or integration with banking/payment systems.
- No fully autonomous business execution or real-time market trading capabilities.
- No guarantee of financial accuracy or investment-level decision approval (AI is advisory only).
- No offline functionality; system requires internet access and API connectivity to LLM services.

8. Assumptions & Constraints

- **LLM Cost Constraint:** The system assumes that an average user session involves multiple agent calls (typically 3–6 agents per query), resulting in an estimated consumption of approximately 2,000–5,000 tokens per session depending on prompt complexity. To manage API cost at scale, the system implements response caching for repeated or similar queries and uses hierarchical summarization so intermediate agent outputs are condensed before being passed to other agents. Additionally, only relevant agents are triggered dynamically instead of running all agents in parallel, significantly reducing unnecessary token usage and inference cost.

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- **Technical Constraints:** The system depends on external LLM APIs (Z.AI GLM, Google Gemini, Anthropic Claude), meaning functionality is constrained by API rate limits, latency variability, and service availability. Document generation is limited to standard export formats (PDF and PPT only) and does not support advanced interactive formats or direct integration with third-party business tools (e.g., CRM systems or financial software). Multi-agent orchestration is handled sequentially in most cases, which may limit real-time parallel reasoning efficiency.
- **Performance Constraints:** Due to multi-agent processing and iterative summarization, complex queries may take approximately 5–15 seconds per response depending on load and token size. To maintain responsiveness, outputs are capped in length (summarized insights instead of full raw analysis per agent), and large-scale reasoning tasks are split into smaller processing steps.
- **User Input:** The system assumes users provide text-based business-related inputs in natural language, with a practical limit of approximately 3,000–8,000 tokens per input depending on model context windows. Inputs exceeding this limit are automatically chunked and summarized. Extremely unstructured or irrelevant inputs (e.g., non-business prompts or raw data dumps without context) may trigger a clarification prompt or partial rejection with guidance for refinement to ensure meaningful AI processing.

9. Risks & Questions Throughout Development

- **Hallucination Risk:** How do we ensure AI-generated business recommendations remain accurate and not misleading?
- **Consistency Across Agents:** How do we maintain alignment when multiple agents produce different perspectives on the same problem?
- **Scalability Risk:** Will API cost and latency remain manageable as user traffic increases?
- **User Dependence:** How do we prevent users from over-relying on AI without critical human judgment?

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- **Input Misinterpretation:** What happens if the system misinterprets vague or incomplete business ideas?
- **Output Trustworthiness:** How do we communicate uncertainty in AI-generated business insights to users effectively?

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